

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & T_z \end{bmatrix}$$

$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & 0 \\ 0 & T_{yy} & 0 \\ 0 & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$

$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{yyz} \\ 0 & 0 & T_{zzz} \\ 0 & T_{yyz} & 0 \\ T_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$

$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$

$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & T_{xxz} \\ 0 & 0 & T_{yyz} \\ 0 & 0 & 0 \\ 0 & -\frac{T_{yyz}}{2} & 0 \\ -\frac{T_{xxz}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxz} = 2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$

$$T_{yyz} = -2\mathbb{M}_{xy}^{(1)} + 2\mathbb{T}_z^{(2)}$$

$$\frac{T_{yyz}}{2} = -\mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$

$$\frac{T_{xxz}}{2} = \mathbb{M}_{xy}^{(1)} + \mathbb{T}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & T_{yzy} & 0 \\ T_{zxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{yzy} = \mathbb{M}_{xy}^{(2)} + \mathbb{T}_z^{(3)}$$

$$T_{zxx} = \mathbb{M}_{xy}^{(2)} - \mathbb{T}_z^{(3)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xyyy} & T_{xxzz} & 0 & 0 & 0 \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & 0 & 0 & 0 \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$

$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$

$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$

$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$

$$T_{zxxx} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$

$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & 0 & 0 \\ -T_{xxyy} & 0 & T_{yyzz} & 0 & 0 & 0 \\ -T_{xxzz} & -T_{yyzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(3)}$$

$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(3)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & 0 \\ 0 & T_{zxxx} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(4)}$$

$$T_{zxxx} = \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 2\mathbb{T}_v^{(4)}$$

$$T_{xyxy} = \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{yzyz} & 0 & 0 \\ 0 & T_{zxxx} & 0 \\ 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(5)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(3)} - 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(3)} + 3\mathbb{T}_u^{(6)} - \mathbb{T}_v^{(6)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(3)} + 2\mathbb{T}_v^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} T_{xxxx} & T_{xyxy} & T_{xxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{yyxx} & T_{yyyy} & T_{yyzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ T_{zzxx} & T_{zzyy} & T_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyyx} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xyxy} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{yyxx} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} + 2\mathbb{T}_v^{(3)}$$

$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzxx} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzyy} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{yzzz} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{zxxx} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(5)}$$

$$T_{xyyx} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & 0 & T_{yzzz} & 0 & 0 \\ 0 & 0 & 0 & 0 & T_{zxzx} & 0 & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{xyxy} & 0 & 0 & T_{xyyx} \end{bmatrix}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)}$$

$$T_{yzzz} = \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} - 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} - 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} - 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)}$$

$$T_{zxxx} = \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} + 2\mathbb{T}_u^{(4)} + 3\mathbb{T}_u^{(6)} + 2\mathbb{T}_v^{(4)} - \mathbb{T}_v^{(6)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(3)} + \mathbb{T}_0^{(3)} + 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)}$$

$$T_{xyyx} = \mathbb{M}_{xyz}^{(3)} - \mathbb{T}_0^{(3)} - 4\mathbb{T}_u^{(4)} + 2\mathbb{T}_v^{(6)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} T_{xxxx} & 0 & 0 \\ 0 & T_{yyyy} & 0 \\ 0 & 0 & T_{zzzz} \\ 0 & 0 & T_{yyzz} \\ T_{zzxx} & 0 & 0 \\ 0 & T_{xxyy} & 0 \\ 0 & T_{yzzz} & 0 \\ 0 & 0 & T_{zzxz} \\ T_{xxyy} + T_{yyzz} - T_{yzzz} - T_{zzxz} + T_{zzxx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T_{xxxx} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{yyyy} = 3\mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{zzzz} = 3\mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzxx} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{yzzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{zzxz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$